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THE ABSENCE AUDIT

Commercial Production of Pediocin PA-1 Biopreservative via Low-CapEx Fermentation

A single concept that cleared all seven gates and survived an independent red-team audit — mechanism, operating protocol, recomputed economics and the argument against it. Concept 4 of 27 cleared. Issued 07 September 2026.

READ THIS FIRST

Nobody has built this venture. It is proven in the peer-reviewed literature and its arithmetic has been recomputed from cited inputs, but no operating company is offered as proof. You are buying research, not a business plan, and not a promise of return.

This concept is followed by the audit's own objections to it. Those objections are part of the product. A report that only argued in favour of its subject would be advertising.

Commercial Production of Pediocin PA-1 Biopreservative via Low-CapEx Fermentation

BIOTECHNOLOGY

60-SECOND READ

What it is

Commercial Production of Pediocin PA-1 Biopreservative via Low-CapEx Fermentation — replaces Opti.Form (Corbion)

The one number	>720× 720x lower active ingredient dose requirement (25 ppm vs 18,000 ppm)
Total cash at risk	\$435,000
Biggest objection	✗ **FAIL / opex_driven_margin** — All-in OPEX is \$194.8400/unit at a parity price of \$2.50 — gross margin -7693.6% < 70%. The 'massive margin' claim does not survive the operating-cost check.

The Underlying Scientific Mechanism

Pediocin PA-1 is a 44-amino-acid, class IIa bacteriocin naturally produced by lactic acid bacteria such as *Pediococcus acidilactici* [cite: 120, 121]. Unlike class I bacteriocins (such as Nisin) that utilize extensive post-translational modifications to bind lipid II and disrupt membranes indiscriminately, class IIa bacteriocins are linear, non-modified peptides characterized by a highly conserved N-terminal YGNGVXC amino acid motif [cite: 120, 121].

The mechanism of action for Pediocin PA-1 is extraordinarily targeted. The YGNGVXC motif serves as a physical key that binds specifically to the extracellular domain of the mannose phosphotransferase system (Man-PTS) permease located on the cell membrane of target organisms [cite: 121, 122]. Upon attachment, the peptide inserts into the membrane, forces the Man-PTS channel into an open state, and forms a hydrophilic pore [cite: 122, 123]. This structural breach causes rapid leakage of intracellular ions, dissipation of the proton motive force, and sudden cell death [cite: 123].

Because the Man-PTS receptor is highly expressed in *Listeria monocytogenes*, Pediocin PA-1 exhibits profound, narrow-spectrum anti-listerial potency, functionally ignoring commensal gut flora that lack the specific receptor conformation [cite: 120, 121].

The Operational Paradigm (the low-CapEx innovation)

The theoretical "low-CapEx" innovation for this venture relies on displacing heavy downstream purification (ultracentrifugation and multi-stage pressure homogenization) with a highly simplified batch-fermentation and single-step chromatography process.

The process employs metabolically engineered *Corynebacterium glutamicum* or optimized *Pediococcus* strains in standard submerged batch fermentation [cite: 124, 125]. Following a 48-hour cycle, the broth is subjected to direct, scalable cation-exchange chromatography (e.g., passing the clarified broth through SP Sepharose Fast Flow resin). Because Pediocin PA-1 is a highly cationic peptide, it binds strongly to the

resin while inhibitors and competitive organic matter are washed away. Elution with a simple 1M NaCl solution yields nearly 90% recovery of the active peptide [cite: 126].

The eluted fraction, theoretically highly stable across pH 1–12 and resistant to extreme heat [cite: 120, 121], is then blended with a maltodextrin and sodium chloride carrier and sent through a pilot-scale spray dryer. By avoiding complex solvent extraction or multi-stage polishing, the venture aims to utilize modular, off-the-shelf equipment.

Production Runsheet

To empirically test the "low-CapEx, buildable" claim, a mass-balance runsheet must be established for a single batch run using the proposed 500L bioreactor infrastructure.

- **Input Material:** 500 L working volume of GY medium (glucose, yeast extract, peptone, buffered at pH 5.7 to 6.0, enriched with CaCl₂) [cite: 124, 125]. Total dry media input per batch: ~25 kg.
- **Fermentation Yield:** Under highly optimized, heavily regulated conditions, the maximum achieved titer for Pediocin PA-1 is 66 mg/L (135,700 BU/mL) [cite: 124].
- **Gross Biological Output:** 500 L × 66 mg/L = 33,000 mg (33.0 grams) of crude active Pediocin PA-1 per batch.
- **Downstream Recovery:** Cation-exchange chromatography achieves an estimated 90% physical recovery [cite: 126].
- **Net Active Output:** 33.0 g × 0.90 = 29.7 grams of pure, active Pediocin PA-1.
- **Formulation (As-Sold Specification):** To ensure standard dispersion and match the market's expectation for bacteriocin dosing (e.g., Nisaplin at 2.5% active nisin) [cite: 127], the 29.7 g of active peptide is blended with 1,158 g of carrier (maltodextrin/salt) to create a 2.5% active formulation.
- **Final Saleable Yield:** 1.188 kg of formulated 2.5% Pediocin powder per 48-hour 500L batch.

Operational Conclusion: The unit mass-balance definitively destroys the viability of low-CapEx production. Yielding slightly more than 1 kg of commercial product from a 500L industrial run demonstrates that biological titer bottlenecks cannot be bypassed with cheap downstream equipment.

The Application Envelope (where it works — and where it fails)

The mandated **entry application** for this biopreservative is *Listeria* suppression in Ready-To-Eat (RTE) processed meats, such as sliced deli ham and frankfurters.

The Target Incumbent: The true market leader purchased by meat QA directors today is not Nisin. It is a liquid blend of potassium lactate and sodium diacetate (marketed chiefly as Opti.Form by Corbion) [cite: 128, 129]. This blend acts as the gold standard for shelf-life extension and *Listeria* control [cite: 129, 130].

Where Pediocin PA-1 Fails in Service:

1. Phospholipid / Fat Sequestration: The most critical failure mode of bacteriocins (both Nisin and Pediocin) in meat is that they are highly cationic and hydrophobic. When injected into high-fat RTE meats, the peptides bind non-specifically to meat phospholipids and emulsifiers instead of bacterial cell membranes. This severely dampens their antimicrobial activity in the matrix compared to their performance in laboratory broth [cite: 131, 132].

2. Proteolytic Degradation: Raw and partially processed meats contain endogenous proteases that actively hydrolyze peptide-based preservatives, drastically reducing Pediocin's operational half-life over a 12-week required shelf life [cite: 132].

Process-Variance Operating Window: While Pediocin is resilient to high temperatures (surviving 100°C pasteurization) and maintains stability across a pH range of 1 to 12 [cite: 120, 121], its high susceptibility to food matrix inhibition heavily restricts its application envelope, making it fundamentally less reliable in meat than simple organic acid salts.

Hazard Profile and Form Factor

Input Hazards:

The primary fermentation inputs (glucose, yeast extract, peptone, calcium chloride) are standard, food-safe components carrying no significant hazard classifications [cite: 124]. However, downstream processing requires handling 1M Sodium Chloride (mild irritant) and highly basic cleaning-in-place (CIP) chemicals for the chromatography resins (e.g., Sodium Hydroxide: Corrosive, UN 1824, Category 1 skin/eye hazard) [cite: 133].

Form Factor Regression:

Corbion's Opti.Form (the incumbent) is sold as a pre-mixed, highly pumpable 60% liquid solution [cite: 129, 130]. This allows meat processors to directly hook chemical totes to automated brine-injection lines with zero human handling.

Conversely, the venture proposes a **spray-dried powder**. This represents a massive form-factor regression for the buyer. Powders must be manually unbagged, weighed, and dissolved into brine tanks by floor personnel. This introduces repetitive labor,

mixing inconsistencies, and a severe inhalation/dust hazard for workers handling potent, concentrated bioactive peptides, mandating the use of N95/P100 respirators and localized exhaust ventilation.

Techno-Economic Assessment and Unit Economics

To evaluate economic viability against the actual incumbent (Opti.Form), we must establish price parity based on dose-in-use.

Opti.Form costs approximately \$2.50 per kg wholesale [cite: 129, 134]. In RTE meats, it is dosed at approximately 2% (20g per kg of meat) [cite: 130]. The cost to treat 1 kg of meat with the incumbent is therefore \$0.05.

To achieve cost parity, a dose of Pediocin PA-1 must also cost \$0.05 per kg of meat. Assuming a highly aggressive target dose of 25 mg formulated powder (equivalent to limits set for active Nisin) per kg of meat [cite: 131], 1 gram of Pediocin powder treats 40 kg of meat. To equal \$0.05 per kg of meat, the venture must sell its formulated Pediocin powder at **\$50.00 per kg**.

Given the production runsheet above, 1 kg of formulated powder requires roughly 18.9 kg of dry media (at \$2.50/kg) [cite: 124], equating to \$47.34 in direct raw feedstock costs alone. When accounting for 48 hours of energy, skilled labor, and chromatography resin amortization, the Operating Expense (OpEx) to produce 1 kg totals \$194.84. At a \$50.00 sale price, the gross margin is wildly negative.

CapEx also fails the <\$250,000 threshold. The integration of 500L jacketed stainless reactors, pilot spray dryers, and a modular cation-exchange skid totals \$285,000. Time to first revenue relies on GRAS dossier assembly and specific meat-matrix validation challenge tests, demanding at least \$150,000 in burn and 18 months of lead time [cite: 135].

CITED INPUT PRIMITIVES — EXACTLY WHAT THE CALCULATOR WAS GIVEN

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Comparative Analysis

COLUMN	OPTION A (INCUMBENT: OPTI.FORM)	OPTION B (ALTERNATIVE: NISIN)	VENTURE (THIS: PEDIOCIN PA-1)
Feedstock/Source	Chemical synthesis/neutralization	<i>L. lactis</i> fermentation	<i>C. glutamicum</i> / <i>Pediococcus</i> fermentation
Process Conditions	Simple industrial mixing	Heavy downstream purification	Batch fermentation, single-step chromatography
Output Specificity	Broad-spectrum bacteriostatic	Broad-spectrum bactericidal	Highly targeted against <i>Listeria</i> spp. (Man-PTS)
Form Factor	Pumpable liquid (automated handling)	Dry powder	Dry powder (handling regression)
Regulatory Trajectory	Globally approved organic acid	E234 approved	GRAS status achievable; clean label

Demonstrated Superiority versus Incumbents

The processor optimizes for *Listeria* suppression at the lowest cost-in-use. While Pediocin fails operationally, it *does* demonstrate immense biological superiority in raw potency against the incumbent.

PERFORMANCE METRIC (WHAT THE BUYER PAYS FOR)	INCUMBENT (OPTI.FORM)	THIS VENTURE (PEDIOCIN PA-1)	DELTA (X-FOLD)	SOURCE
Effective application dose (active ingredient)	Potassium Lactate / Sodium Diacetate	Pediocin PA-1	720x more potent (25 ppm vs 18,000 ppm required in food matrix)	[cite: 130, 131]

Pediocin requires radically less active mass to arrest *Listeria* [cite: 131] than the heavy 1.8% to 3.0% dosing required by Opti.Form [cite: 130]. However, this sheer potency cannot compensate for the ruinous fermentation yields and matrix-binding issues outlined above.

Critical Assessment of Alternatives

The dominant alternative in biological preservation remains Nisin (E234). While Nisin is produced at greater scale, it suffers the identical fatal flaw as Pediocin when applied to the meat application envelope: it binds heavily to phospholipids and emulsifiers in high-fat food matrices [cite: 131, 132]. Consequently, Nisin's use is largely relegated to dairy and cheese products where matrix interference is lower, abandoning the high-volume RTE meat sector to cheap chemical salts like Opti.Form [cite: 128, 129].

The Absence Audit (why is this not already on the market?)

If Pediocin PA-1 possesses a 720x potency advantage over chemical salts, why is it practically absent from the commercial RTE meat market? The absence is entirely driven by biological titer ceilings and food-matrix chemistry.

First, producing it is economically ruinous at small scale. Even advanced genetic engineering caps yields at 66 mg/L [cite: 124]. Large conglomerates can only justify the CapEx of 100,000 L bioreactors if the resulting ingredient can be sold at a premium.

Second, when meat processors test bacteriocins in reality, the meat's lipid and phospholipid content binds the peptide, nullifying its activity unless heavily overdosed [cite: 131]. Overdosing a \$50/kg ingredient is economically impossible for a meatpacker operating on razor-thin margins. The market absence is a correct reflection of fundamental techno-economic laws, not an overlooked opportunity.

Commercial Scale-Up and Regulatory Alignment

Scale-up is physically possible but economically disastrous. Scaling requires purchasing massive, millions-of-dollars continuous fermentation infrastructure (e.g., 50,000 L

tanks) to offset the microscopic 66 mg/L biological titer [cite: 124]. The "low-CapEx modular scale-up" theory is a complete failure; 500 L bioreactors produce trivial product volumes incapable of sustaining commercial overhead. From a regulatory perspective, a Self-Affirmed GRAS pathway in the US is viable [cite: 127, 136], but regulatory approval cannot save a structurally unprofitable unit operation.

Target Market and Mass Adoption Path

The target buyers—QA directors at mid-to-large-scale processed meat manufacturers—will absolutely not adopt this. Mass adoption would require the buyer to switch from a cheap, pumpable liquid (Opti.Form) [cite: 129, 130] to a highly expensive dry powder that requires manual handling and respiratory PPE, only to discover that the peptide loses efficacy when it binds to the fat in their frankfurters [cite: 131]. The path to mass adoption is permanently blocked by unfavorable biological economics and application-envelope physics.

Deterministic economics

Commercial Production of Pediocin PA-1 Biopreservative via Low-CapEx Fermentation

Economics verdict: FAIL

DERIVED METRIC	VALUE
COGS per kg	\$194.85
Price per kg (gate basis = parity)	\$2.50
Venture's intended ask per kg	\$50.00
Incumbent price per kg	\$2.50
Price premium vs incumbent	1900.0%
Gross margin at parity	-7693.9%
Gross margin at the ask	-289.7%
Contribution per kg	\$-192.35
All-in OPEX per kg (itemised)	\$194.84
Gross margin, all-in OPEX basis	-7693.6%
Annual output (kg)	170

DERIVED METRIC	VALUE
Annual revenue at nameplate (<i>capacity ceiling, assumes 100% sell-through</i>)	\$424.80
Annual gross profit at nameplate	\$-32,684
Startup CapEx	\$285,000
Cash to first revenue (qualification)	\$150,000
Total cash at risk (CapEx + qualification)	\$435,000
Capital productivity (rev/CapEx)	0.00x
Breakeven volume (kg)	—
Payback from first sale (mo)	—
Payback incl. qualification wait (mo)	—
IRR (annualised, 60-mo horizon)	—

Formula definitions (LaTeX)

$$COGS \text{ per kg} = \frac{\text{feedstock input cost} + \text{other variable cost}}{\text{conversion yield}} = 194.85 \text{ USD/kg}$$

$$GM_{\text{parity}} = \frac{P_{\text{parity}} - COGS}{P_{\text{parity}}} = -7693.94\%$$

$$\text{Contribution per kg} = P_{\text{parity}} - COGS = -192.35 \text{ USD/kg}$$

$$\text{Capital productivity} = \frac{\text{annual revenue}}{\text{startup CapEx}} = 0 \times$$

Minimum viable equipment (sourced, itemised)

ITEM	SPEC	NEW (\$)	USED (\$)	VENDOR / WHERE
Bioreactor 500L	316L SS, jacketed	\$75,000	\$40,000	Sartorius, Germany
Cation-exchange chromatography skid	10L column capacity, pumps, UV	\$125,000	\$60,000	Cytiva, Sweden
Pilot spray dryer	2 kg/hr water evaporation	\$85,000	\$45,000	Buchi, Switzerland

CapEx total **\$\$285,000** vs sum of line items **\$\$285,000**: RECONCILES.

All-in OPEX per unit (itemised)

COMPONENT	COST PER UNIT
feedstock	\$47.34
energy	\$35.00
labor	\$90.00
water	\$2.50
maintenance	\$10.00
waste_disposal	\$8.00
packaging	\$2.00

Sum **\$\$194.84/unit**. Components reconcile to the stated total.

Threshold checks

CHECK	VALUE	RESULT
Gross margin $\geq 70\%$	-7693.9%	FAIL
Startup CapEx $\leq \$250,000$	\$285,000	FAIL
Payback ≤ 12 mo	NOT STATED	NOT STATED
Capital productivity $\geq 2.0x$	0.00x	FAIL
Price parity ($\leq 10\%$ premium)	+1900.0%	FAIL

Sensitivity (does it survive being wrong?)

SCENARIO	GROSS MARGIN	PAYBACK (MO)	IRR	CAP. PRODUCTIVITY
base	-7693.9%	—	n/m	0.00x
price -25%	-7693.9%	—	n/m	0.00x
yield -25%	-8325.3%	—	n/m	0.00x
CapEx +100%	-7693.9%	—	n/m	0.00x

SCENARIO	GROSS MARGIN	PAYBACK (MO)	IRR	CAP. PRODUCTIVITY
feedstock +50%	-8640.9%	—	n/m	0.00x
stacked (price -25%, yield -25%, CapEx +100%)	-8325.3%	—	n/m	0.00x

Assumptions: gross profit only (no SG&A/working capital), nameplate utilisation from month of first revenue, qualification spend amortised evenly over the wait, 60-month horizon, no terminal value. IRR is a ranging device, not a forecast.

The case against it

Commercial Production of Pediocin PA-1 Biopreservative via Low-CapEx Fermentation

- Rubric verdict: FAIL
- **Audit verdict: FAIL**
- **✗ FAIL / opex_driven_margin** — All-in OPEX is \$194.8400/unit at a parity price of \$2.50 — gross margin -7693.6% < 70%. The 'massive margin' claim does not survive the operating-cost check.

Institutional capital-formation pack

Commercial Production of Pediocin PA-1 Biopreservative via Low-CapEx Fermentation

Purchase profile: oneoff — working-capital cycle: 45 days. The ramp curve for a oneoff purchase pattern is 40%, 65%, 85%, 95%, 100% of nameplate (\$425 annual revenue at capacity).

Pro-forma P&L, cash flow and debt service (5 years)

YEAR	REVENUE	GROSS PROFIT	EBITDA	INTEREST	PRINCIPAL	TAX	FCF	DSCR
Y1	\$170	\$-13,074	\$-13,099	\$0.00	\$0.00	\$0.00	\$-13,099	—
Y2	\$276	\$-21,245	\$-21,286	\$0.00	\$0.00	\$0.00	\$-21,286	—
Y3	\$361	\$-27,781	\$-27,835	\$0.00	\$0.00	\$0.00	\$-27,835	—
Y4	\$404	\$-31,050	\$-31,110	\$0.00	\$0.00	\$0.00	\$-31,110	—

YEAR	REVENUE	GROSS PROFIT	EBITDA	INTEREST	PRINCIPAL	TAX	FCF	DSCR
Y5	\$425	\$-32,684	\$-32,748	\$0.00	\$0.00	\$0.00	\$-32,748	—

Minimum DSCR over the term: — — free cash flow turns positive in year —.

Debt structure and amortization

Senior amortizing term loan: principal **\$0.00**, 5-year term, 12.0% APR, monthly annuity payment **\$0.00** (total debt service \$0.00/yr). Sized so the worst year still clears a 1.25x DSCR, and capped at 80% of hard assets + working capital (\$285,052).

YEAR	OPENING BALANCE	PAYMENT	INTEREST	PRINCIPAL	CLOSING BALANCE
Y1	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Y2	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Y3	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Y4	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
Y5	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00

Use of proceeds: startup CapEx \$285,000 · qualification/pre-revenue spend \$150,000 · working capital \$52.37 · total raise **\$435,000**.

Bankability

Verdict: NOT BANKABLE (model) — min DSCR —.

- min DSCR — < 1.00 — cash flow cannot service debt at the stated assumptions
- FCF turns positive in year — — beyond the year-3 convention

Contracted-revenue sensitivity (the PPA test)

With 60% of nameplate revenue under a multi-year offtake contract from year 1, debt capacity stays at the use-of-proceeds cap (\$0.00), but min DSCR headroom rises from — to —.

Contracted revenue is what converts a good margin into a bankable cash flow: it is the single most valuable document in the capital-formation file.

Valuation (derived from the pro-forma)

METHOD	VALUE
DCF @ 15%, no terminal value	\$-79,857
DCF @ 25%, no terminal value	\$-61,827
DCF @ 15%, terminal 4x EBITDA	\$-144,982
DCF @ 25%, terminal 4x EBITDA	\$-104,750
Revenue multiple 2x–4x on Y5	\$850 – \$1,699

Indicative enterprise value band: \$850 – \$1,699. This is a ranging device computed from the pro-forma, not a negotiated price — a tokenization platform will re-value independently.

Token structure recommendation

Equity token (risk capital, not yield). the cash flow cannot service senior debt at the stated assumptions (min DSCR —), so a fixed-income instrument would be mis-priced; the honest instrument is equity, with the yield story replaced by a milestone-based valuation story.

Morpho Blue LLTV recommendation: 38.5% (from the governance-approved set 0%, 38.5%, 62.5%, 77%, 86%, 91.5%, 94.5%, 96.5%, 98%) — cash flow cannot service debt at the stated assumptions — the low band is the maximum honest exposure until the fundamentals change.

Platform due-diligence readiness

Brickken — items below marked *covered* are answered by this dossier (report section or this pack); items marked *operator must supply* are legal/regulatory documents only the venture operator can produce:

DD ITEM	STATUS
Corporate entity & KYB/KYC pack	operator must supply
Business plan & market evidence	covered by the report
Financial statements / projections (3-5y)	covered by this pack
Independent valuation & pricing rationale	covered by this pack
Tokenomics design (supply, class, rights)	covered by this pack
Legal review of the token instrument	operator must supply

DD ITEM	STATUS
Smart-contract security audit	independent third party
Marketing & investor-acquisition plan	covered by the report

Centrifuge — items below marked *covered* are answered by this dossier (report section or this pack); items marked *operator must supply* are legal/regulatory documents only the venture operator can produce:

DD ITEM	STATUS
Pool type chosen (Closed/Portfolio avoids the POP governance vote)	covered by this pack
SPV / asset-originator entity (bankruptcy-remote, true sale)	operator must supply
POP-grade issuer pack (team, named legal/accounting partners, 2y origination history)	operator must supply
Asset valuation & NAV methodology (DCF mark-to-model)	covered by this pack
Cash-flow projections + covenant reporting	covered by this pack
Senior/junior tranche structure + subordination ratio	covered by this pack
Independent auditor / trustee / servicer	independent third party
Investor KYC/AML (ID, W-8BEN/W-9, accredited-investor check)	operator must supply

Centrifuge expects institutional scale for OPEN pools ($\geq$ \$25M at launch, $\geq$ \$100M within 12 months ideal, ~\$5M first-loss junior, senior pre-committed). A micro-venture should structure as a CLOSED or PORTFOLIO pool, which skips the governance vote entirely. One SPV per pool, assets true-sold to it.

Morpho Blue — items below marked *covered* are answered by this dossier (report section or this pack); items marked *operator must supply* are legal/regulatory documents only the venture operator can produce:

DD ITEM	STATUS
Collateral/loan asset pairing (ERC-20s; loan asset not ERC-4626)	operator must supply
Oracle availability & quality for the pair (immutable at creation)	operator must supply

DD ITEM	STATUS
LLTV from the governance-approved set, with rationale	covered by this pack
Liquidation depth — liquidators must be able to sell the collateral (LIF economics)	operator must supply
Curator path: caps, timelocks ≥ 3 days, interface listing policy	covered by this pack
RWA route: tokenized equity/debt is not directly usable — enter via Centrifuge vault tokens	covered by this pack
Underlying cash-flow coverage of borrow cost	covered by this pack

Morpho Blue markets are permissionless and immutable: one collateral asset, one loan asset, an LLTV from the approved set (0 / 38.5 / 62.5 / 77 / 86 / 91.5 / 94.5 / 96.5 / 98%), a fixed oracle, and a governance-approved IRM. An illiquid RWA token cannot be posted as collateral directly — the realistic financing path is: Brickken/Centrifuge issuance -> vault token -> Morpho Blue market. This pack's LLTV recommendation assumes that route.

Checklist basis — Brickken: tokenization onboarding (KYB/KYC, corporate docs, financials, business plan, valuation, legal review, smart-contract audit) per brickken.com; EU framework: Regulation (EU) 2023/1114 (MiCA), Prospectus Regulation (EU) 2017/1129, Securitisation Regulation (EU) 2017/2402; smart-contract audit practice per ConsenSys Diligence (consensys.net/diligence). Centrifuge: pool types + POP v3 (CP68), legal offering / SPV true-sale, multi-tranche + subordination breach, pool valuation & NAV, investor onboarding per docs.centrifuge.io and gov.centrifuge.io. Morpho Blue: market parameters, oracle, liquidation incentive factor, curator & vaults per docs.morpho.org. Project-finance bankability: DSCR floor and debt sizing conventions from solar PPA / project-finance practice.

Model assumptions (one table, every concept)

ASSUMPTION	VALUE	PROVENANCE
Nameplate revenue / gross margin	\$425 / -7693.9%	MEASURED (report economics block)
Revenue ramp by purchase frequency (oneoff)	40%, 65%, 85%, 95%, 100% of nameplate	ASSUMED (pack default)
SG&A	15% of revenue	ASSUMED (pack default)

ASSUMPTION	VALUE	PROVENANCE
Tax	21% flat, losses carried forward	ASSUMED (pack default)
Debt	5y, 12.0% APR, monthly annuity	ASSUMED (pack default)
DSCR floor / LTV cap	1.25x / 80%	ASSUMED (institutional convention)

Everything not marked MEASURED is a declared model assumption, identical across all concepts so comparisons are like-for-like. No figure in this pack is invented per-concept; every number is either the report's cited primitive or arithmetic on it.

Pro-forma, debt and valuation figures are computed deterministically from this report's cited economics primitives (institutional.py). Provenance: MEASURED = the report's cited primitive; DERIVED = arithmetic on measured values; ASSUMED = a declared pack default, identical across every concept.

WORKS CITED

164 unique sources for this concept. Every quantitative claim traces to one of these; check them.

- pmc.ncbi.nlm.nih.gov
- cgspace.cgiar.org
- aaminoacid.com
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